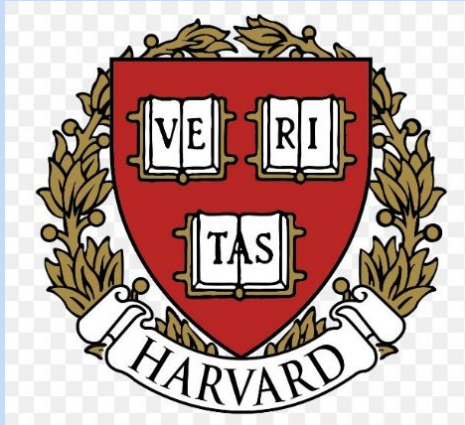


Bio-Bootcamp #1

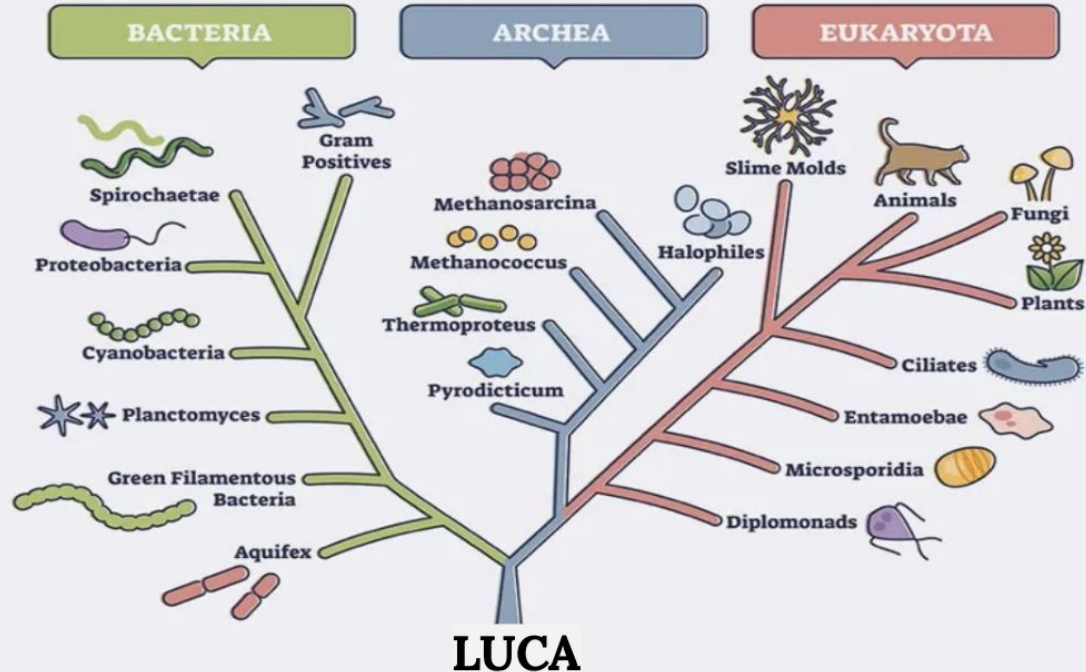
The Central Dogma of Biology and its Applications in Synthetic Biology

Anna-Thérèse Mehra
Harvard College '24



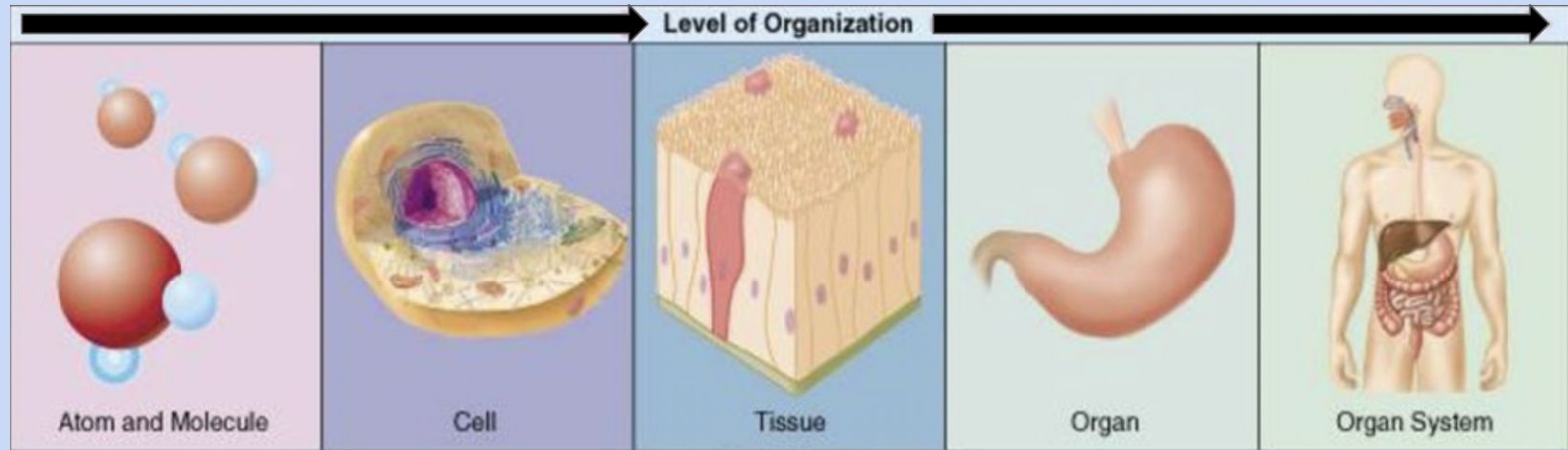
Evolutionary History of Life

PHYLOGENETIC TREE

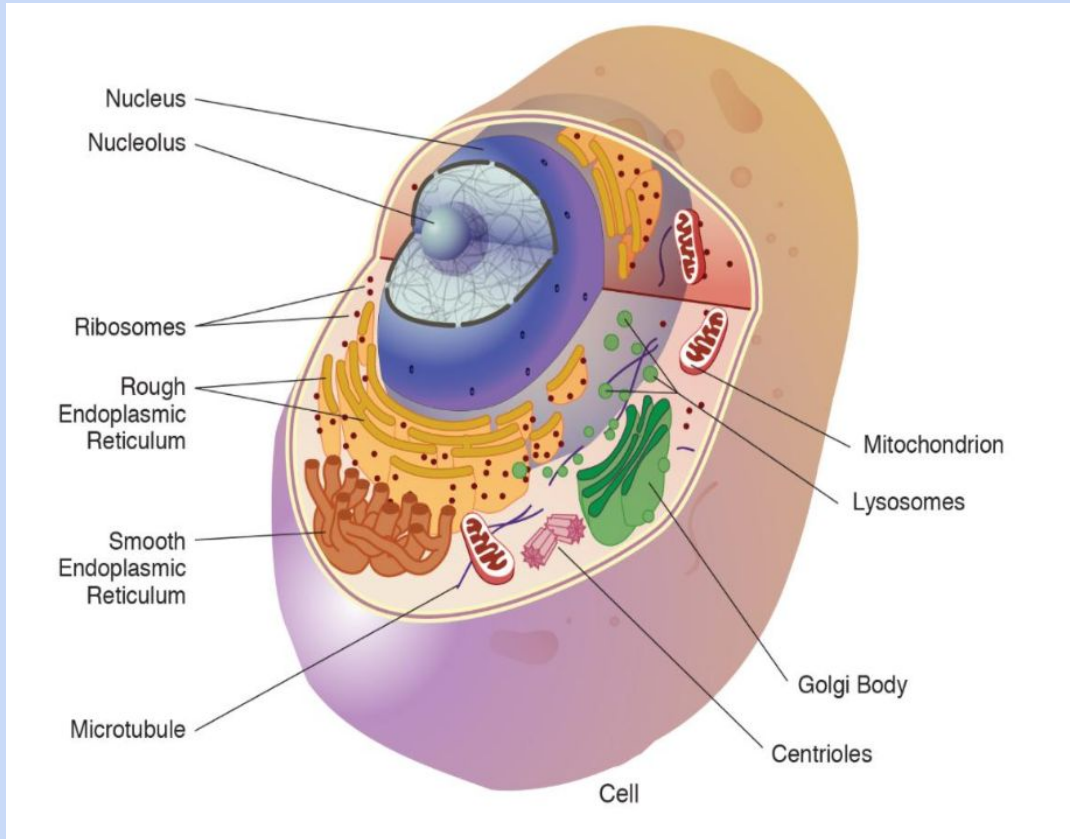


The Organized Human Body

Layers of Organization From Simple to Complex

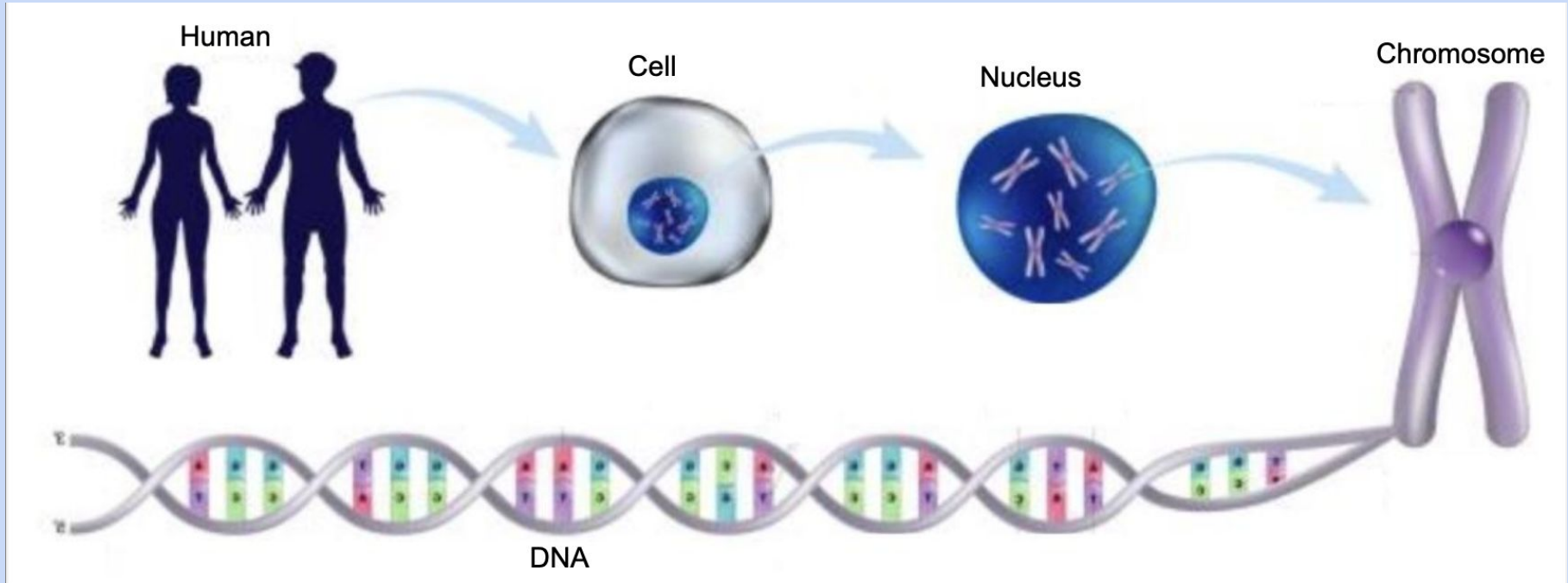


Cell Organelles



Trillions of Cells Carry the DNA

Human DNA is compacted into 46 chromosomes



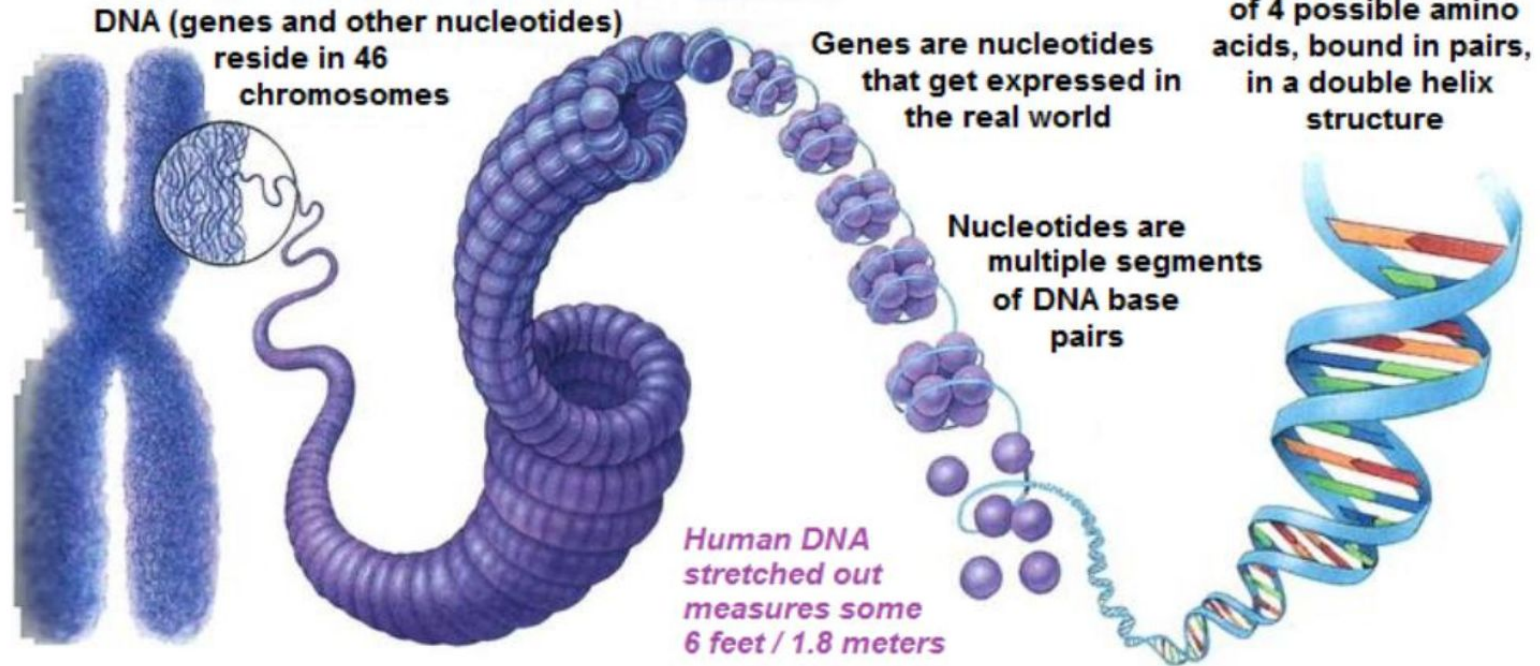
The Nucleus: the Keeper of DNA



DNA Unravelled:

From Chromosome to Double Helix

The Hierarchical Structure of DNA through to the Chromosome



Functions of DNA

DNA Replication and Protein Synthesis

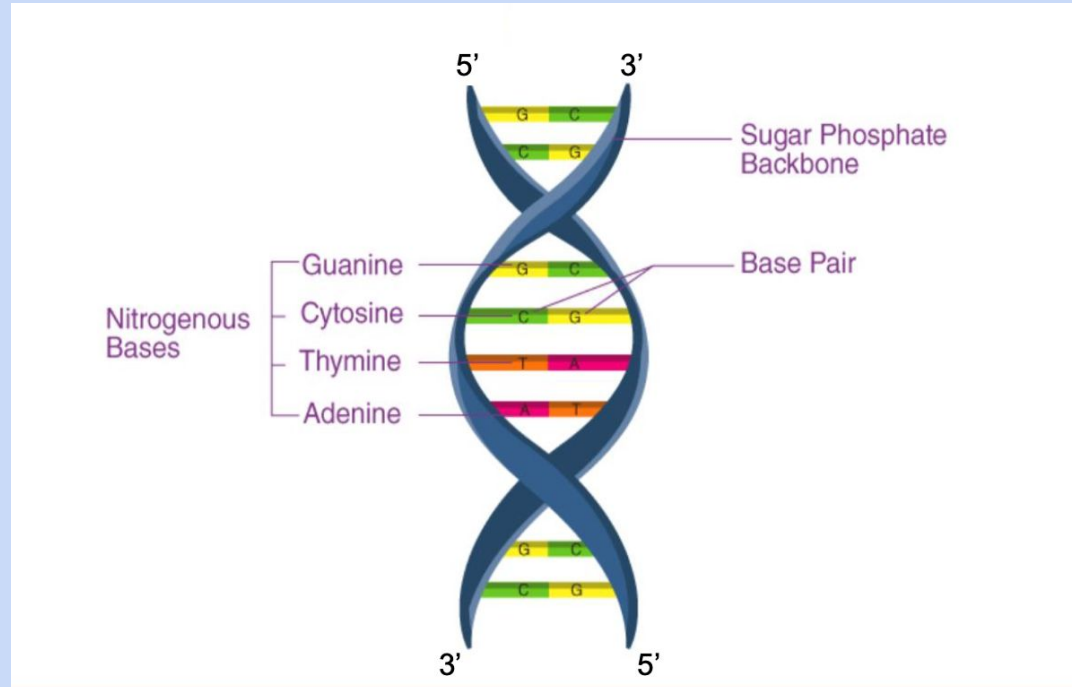
DNA Replication

- Produces DNA
- Ensures that each new cell receives complete set of DNA identical to parent cell
- Occurs during S phase (Synthesis phase) of cell cycle

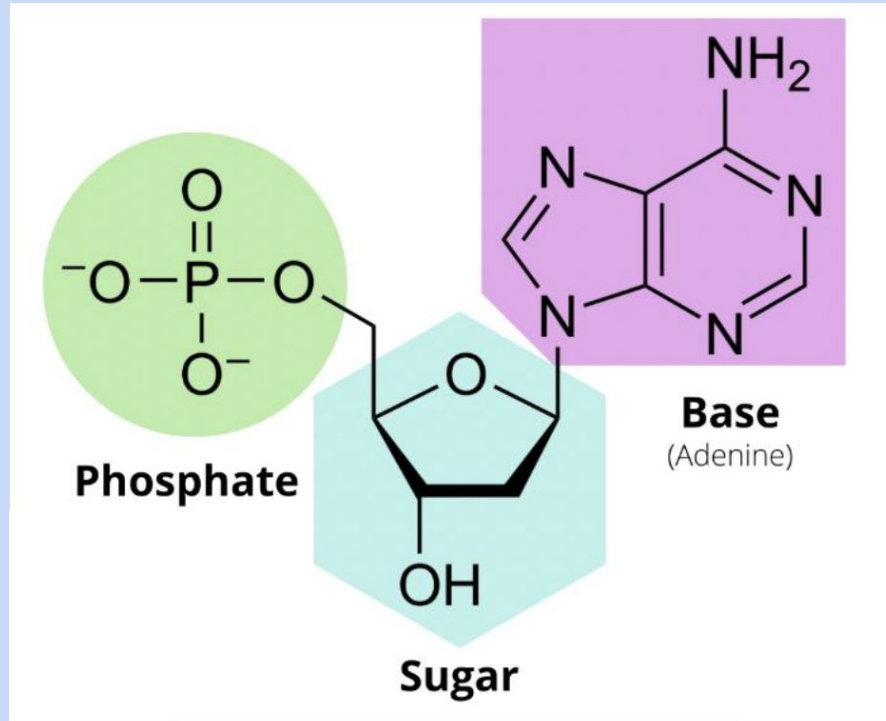
Protein Synthesis

- Produces proteins
- DNA provides blueprint for building proteins vital for cell function
- Occurs during G1 and G2 phases of cell cycle, resulting in cell growth and division

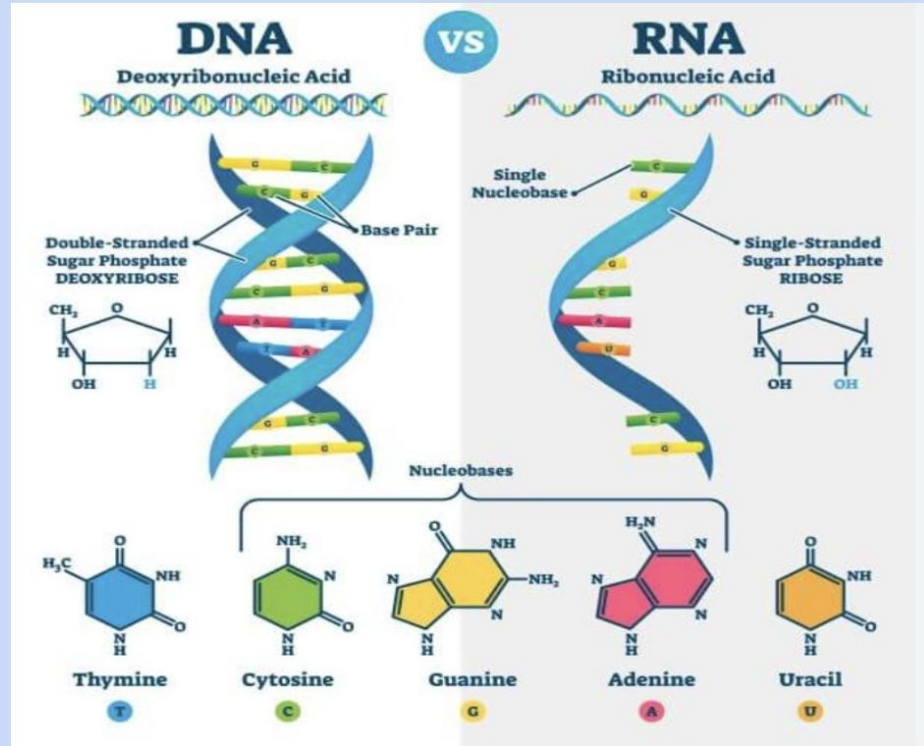
DNA Double Helix



Structure of Nucleotide



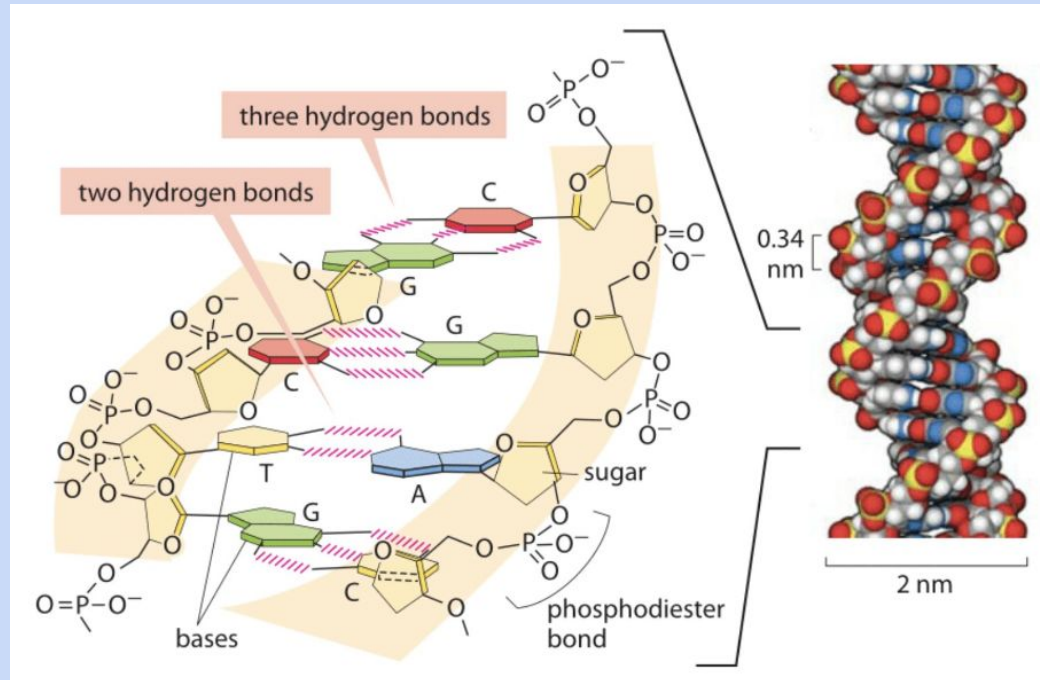
Structure of Nucleic Acids



DNA Double Helix Bonding

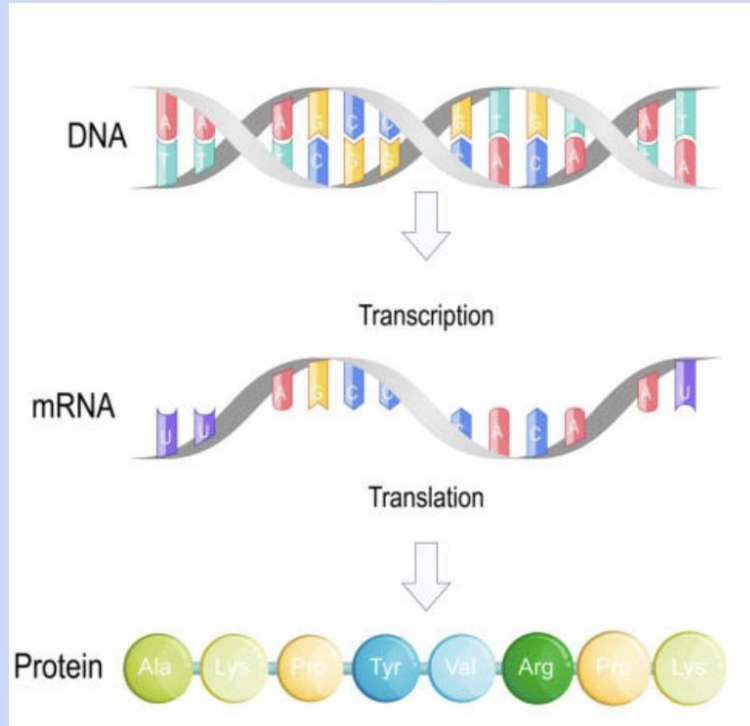
Hydrogen Bonds Between Bases

Phosphodiester Bonds of Backbone



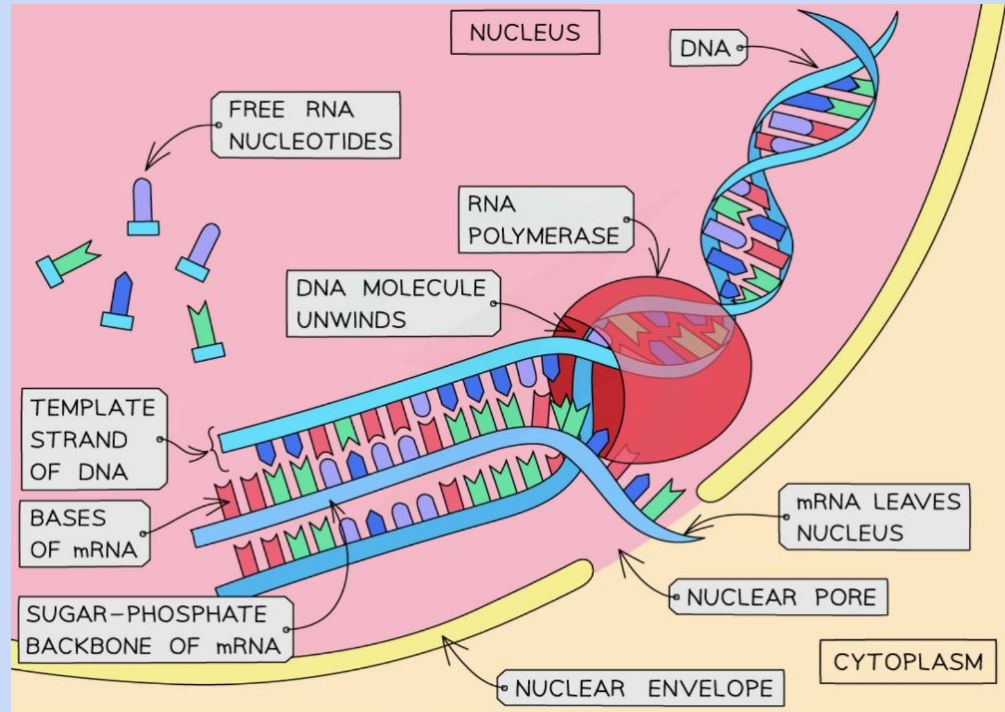
Overview of Protein Synthesis

DNA → mRNA → Protein



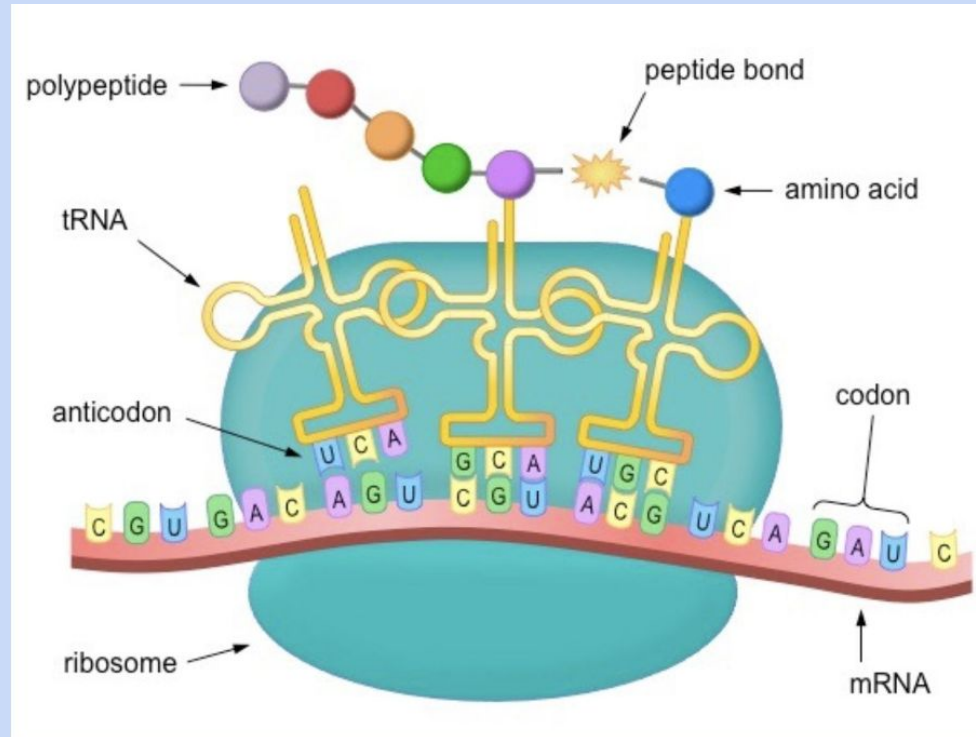
DNA Transcription:

DNA → mRNA

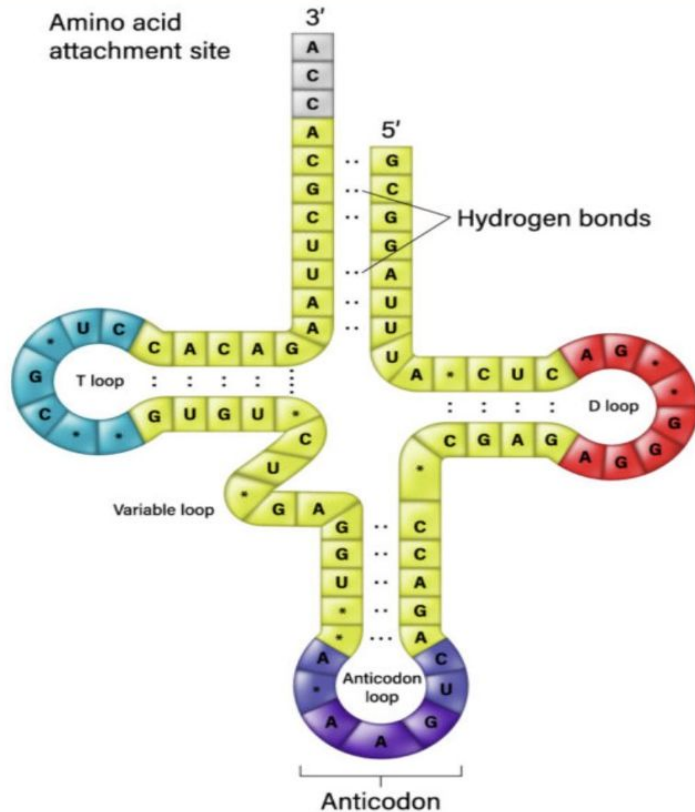


mRNA Translation:

RNA → Protein



tRNA Molecule

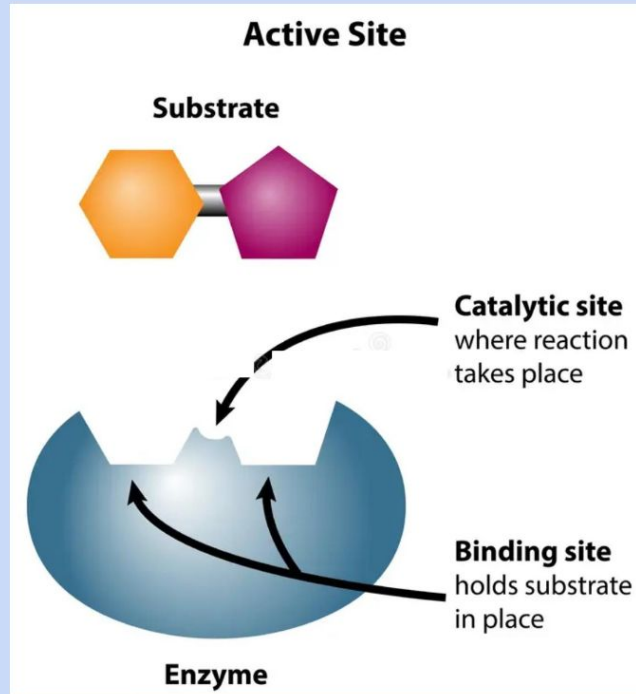


The Genetic Code

		Second nucleotide				
		U	C	A	G	
First nucleotide	U	UUU Phe UUC UUA Leu UUG	UCU UCC Ser UCA UCG	UAU Tyr UAC UAA STOP UAG STOP	UGU Cys UGC UGA STOP UGG Trp	U C A G
	C	CUU CUC Leu CUA CUG	CCU CCC Pro CCA CCG	CAU His CAC CAA Gln CAG	CGU CGC Arg CGA CGG	U C A G
	A	AUU Ile AUC AUA AUG Met	ACU ACC Thr ACA ACG	AAU Asn AAC AAA Lys AAG	AGU Ser AGC AGA Arg AGG	U C A G
	G	GUU GUC Val GUA GUG	GCU GCC Ala GCA GCG	GAU Asp GAC GAA Glu GAG	GGU GGC Gly GGA GGG	U C A G

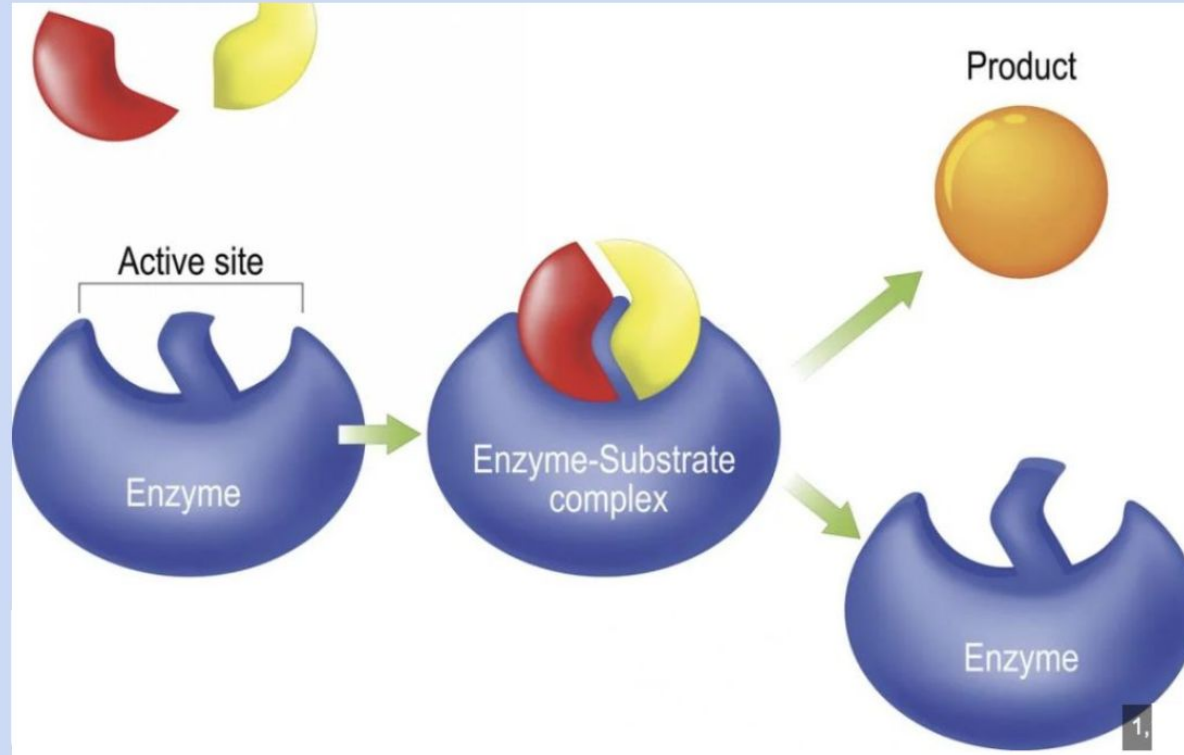
Enzymes are Proteins

Enzyme Structure



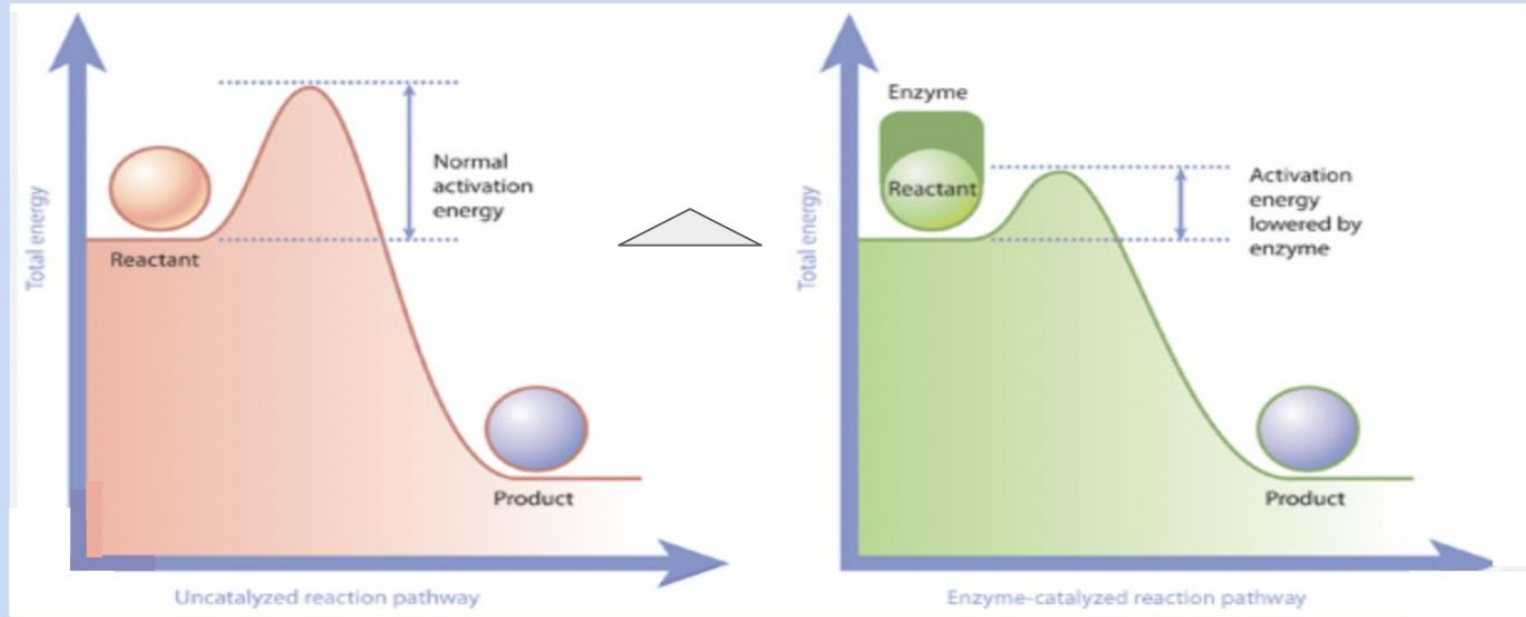
Function of Enzymes

Enzymes convert substrate molecules into product molecules



Function of Enzymes

Enzymes speed reaction rate by lowering “activation energy”

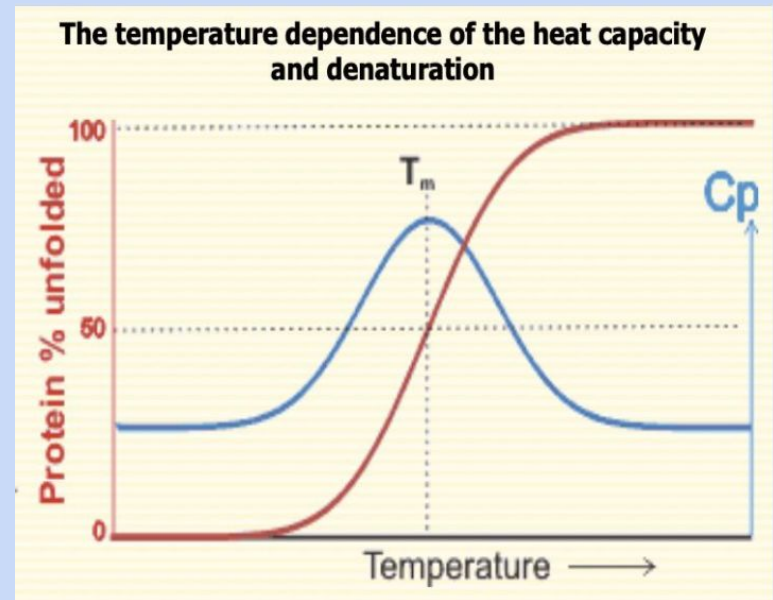
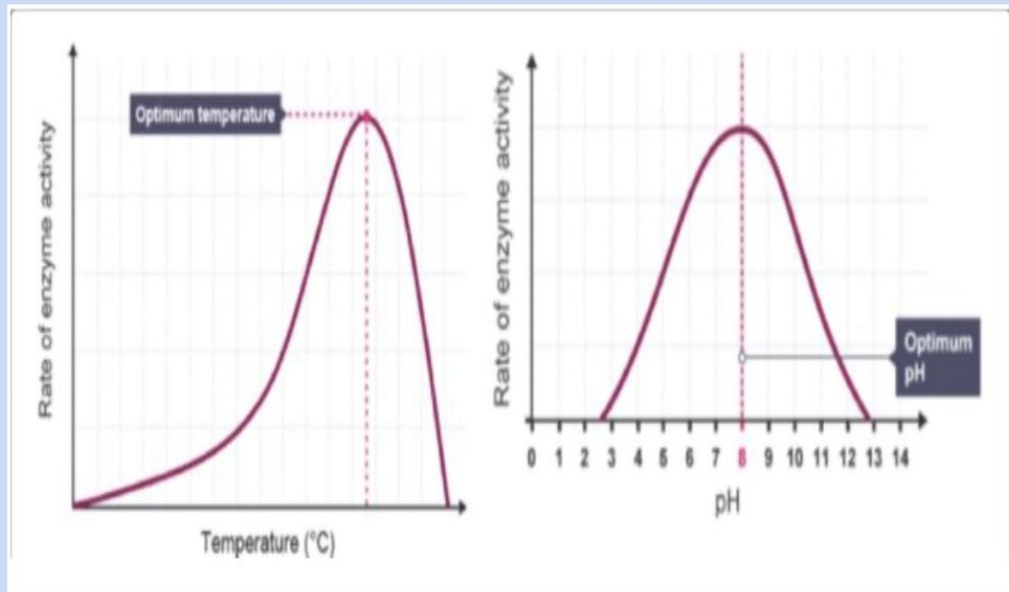


Function of Enzymes

Enzymes can be denatured

Lose function when exposed to suboptimal pH and/or excessive heat

Ensures specificity of enzyme's unique 3-D structure



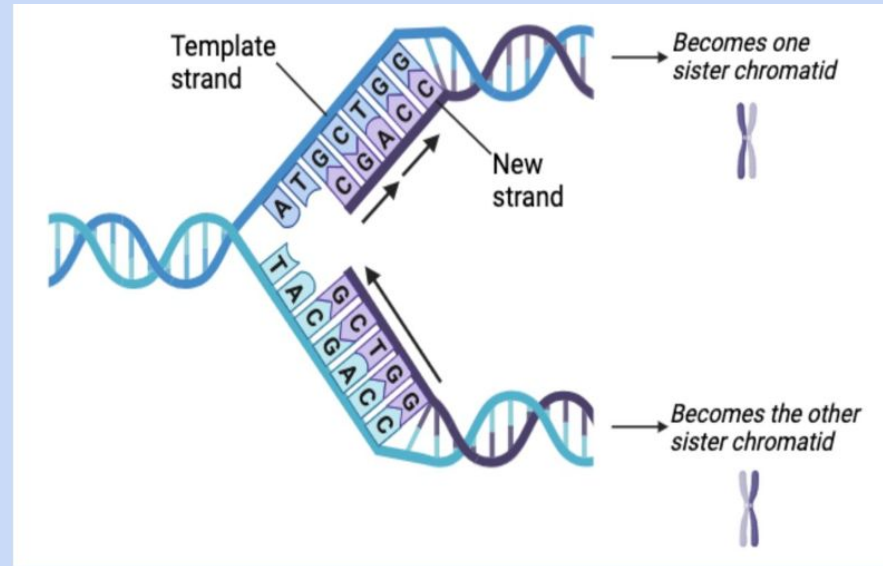
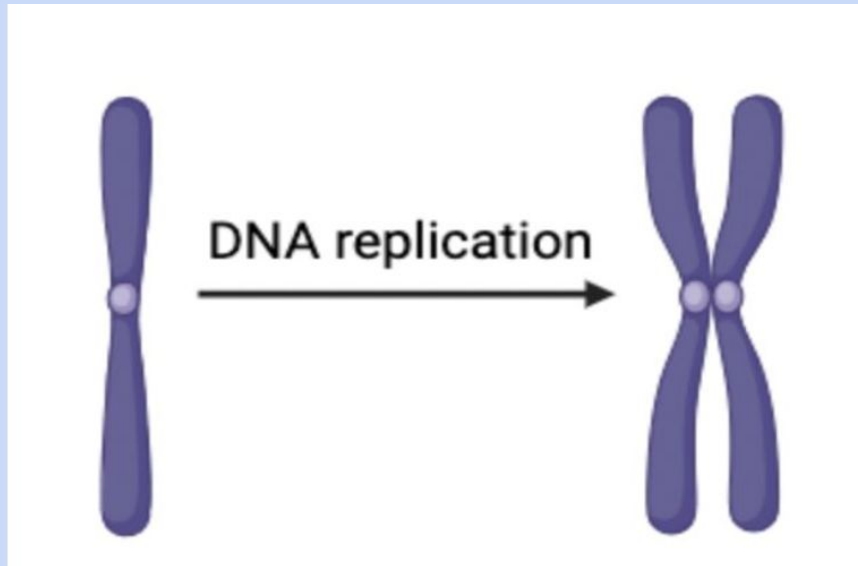
“I’m an enzyme, please keep me frozen!”



DNA Replication

Each chromosome is copied, forming pair of sister chromatids

Pair gets divided into two daughter cells

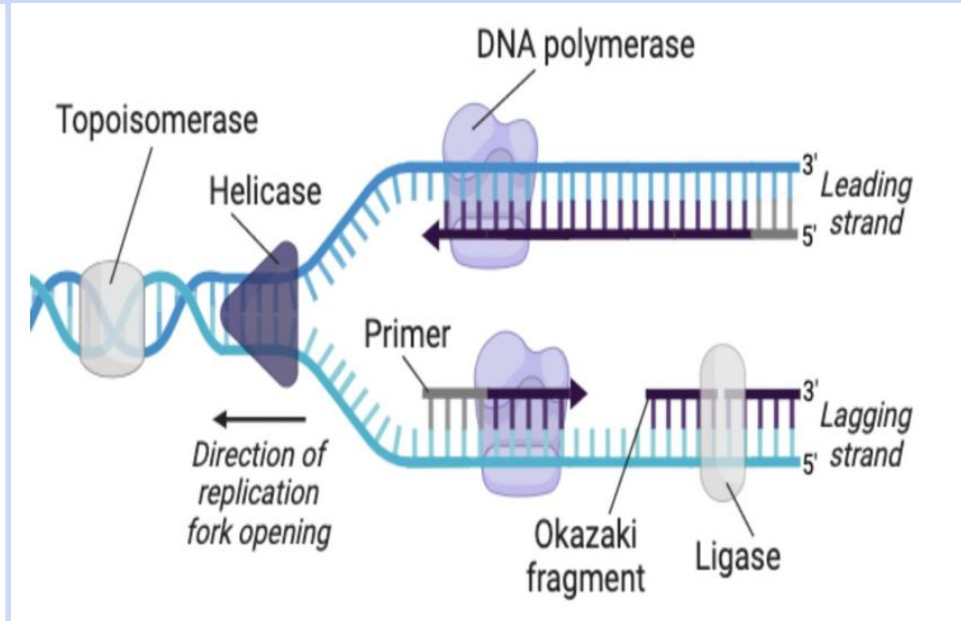
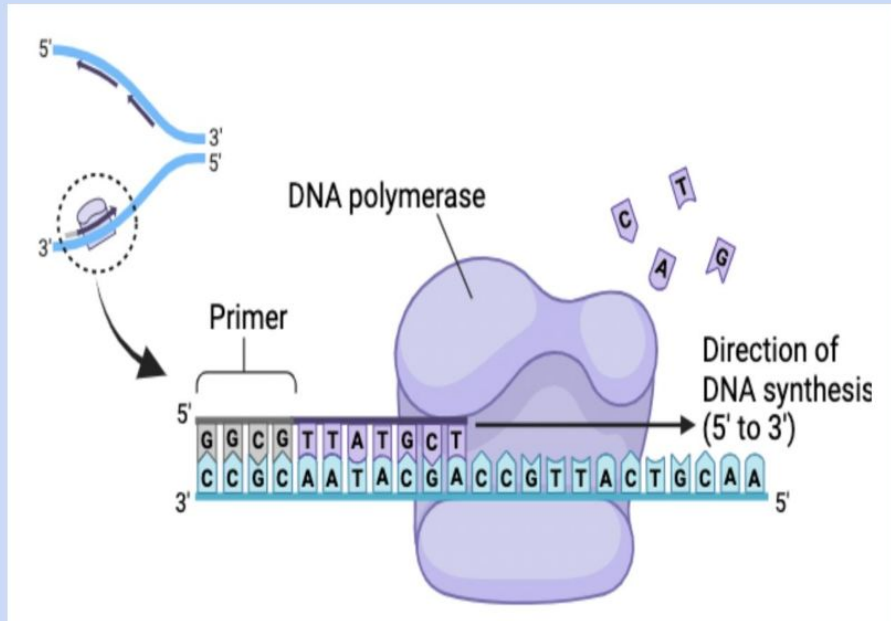


DNA Replication

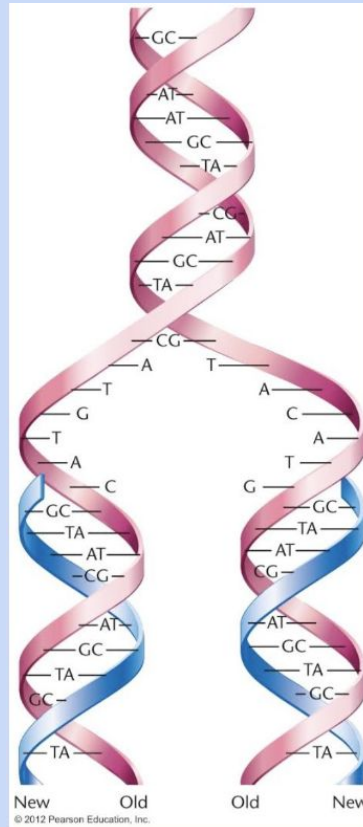
Helicase (enzyme) unravels the double helix

Topoisomerase (enzyme) cuts and rejoins DNA strands

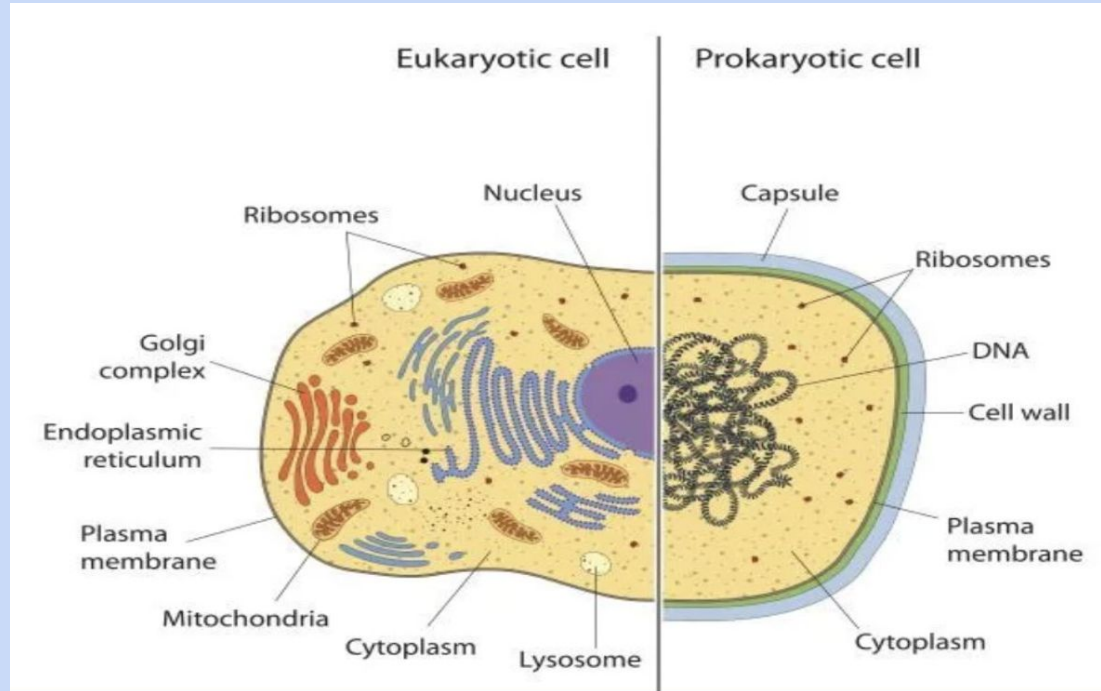
DNA Polymerase (enzyme) assembles nucleotides into a new DNA strand



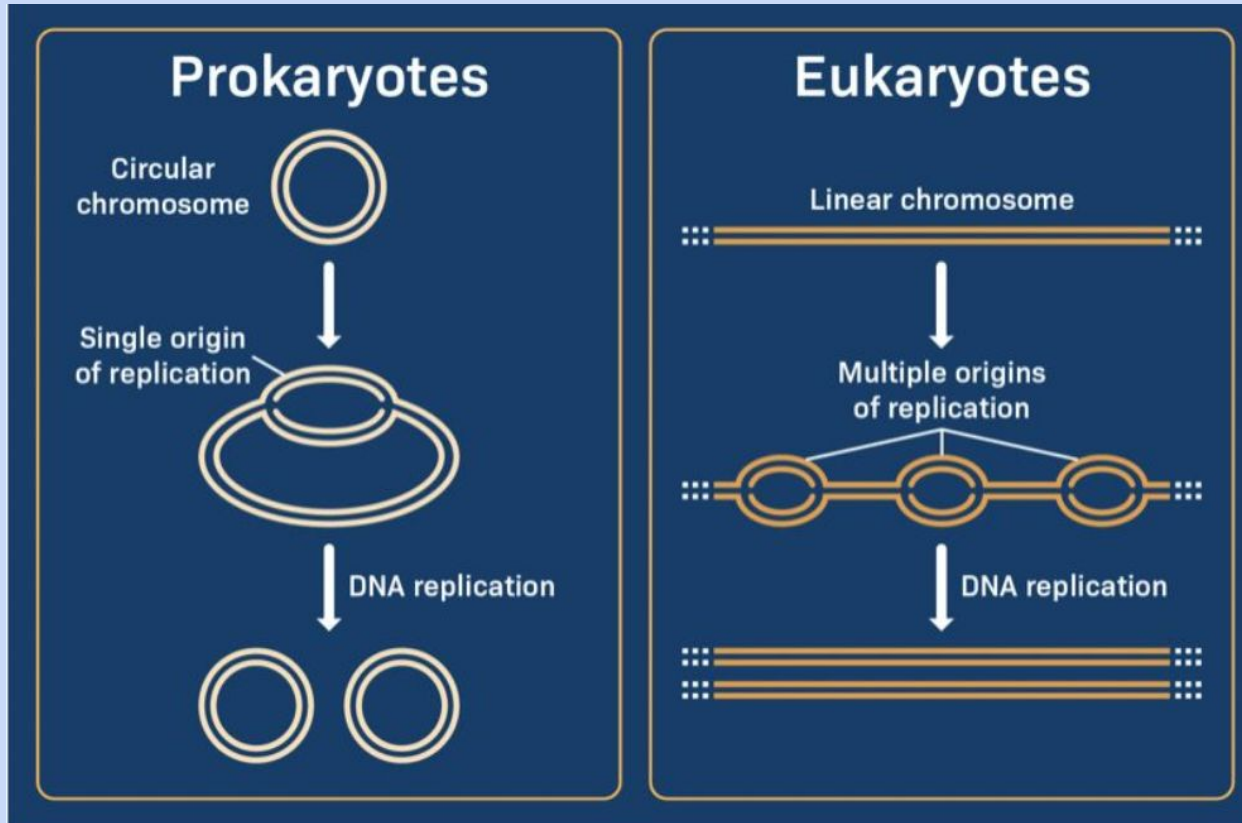
DNA Replication: Semi-conservative Model



Prokaryotic vs. Eukaryotic Cell Organization

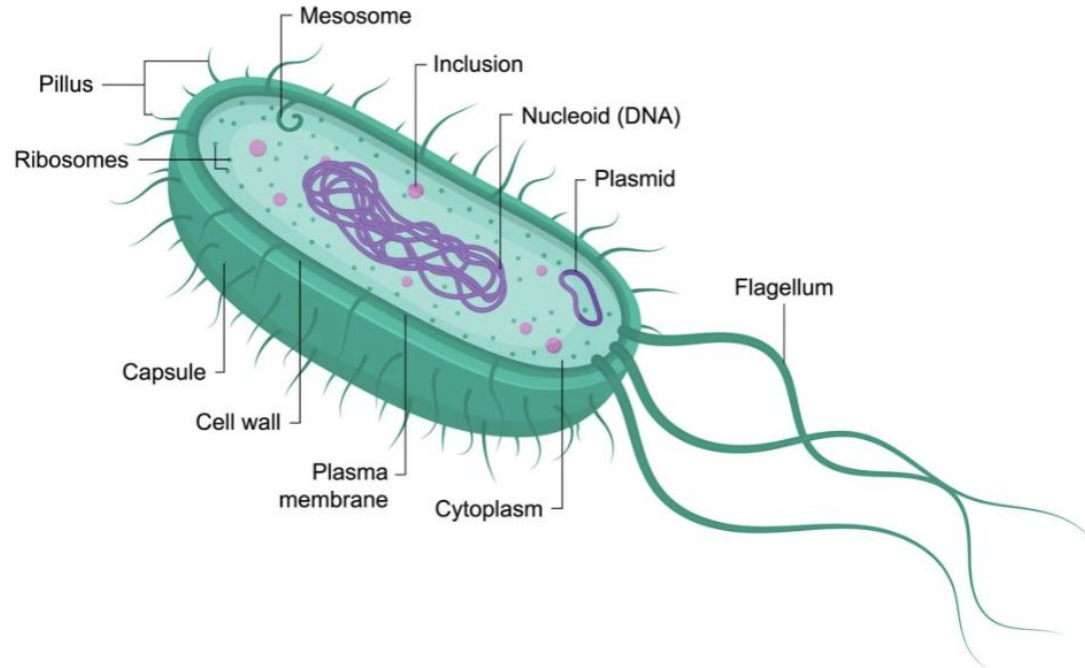


Prokaryotic vs. Eukaryotic DNA Replication



Bacterial Structure

BACTERIAL CELL



How does PCR modify natural tools?

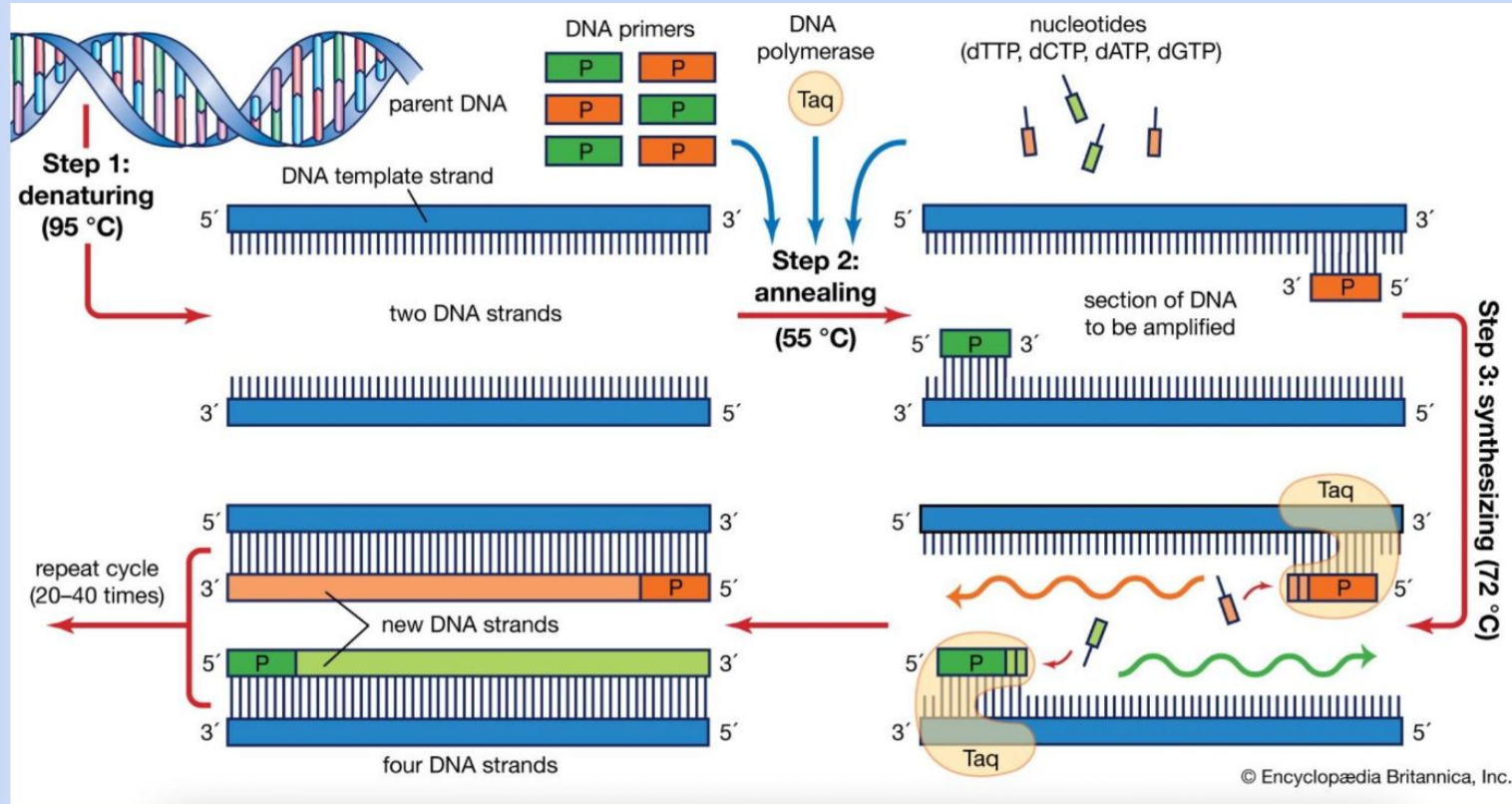
- Mimics DNA replication, in order to generate millions of copies of DNA pieces
- PCR amplifies targeted segments of DNA, while DNA replication is undergone by all of a cell's DNA
- PCR replaces most of the enzymes needed in replication with heat, except for DNA Polymerase

Substrate = DNA Template

Enzyme = Taq Polymerase

Product = Millions of copies of DNA pieces

Polymerase Chain Reaction (PCR)

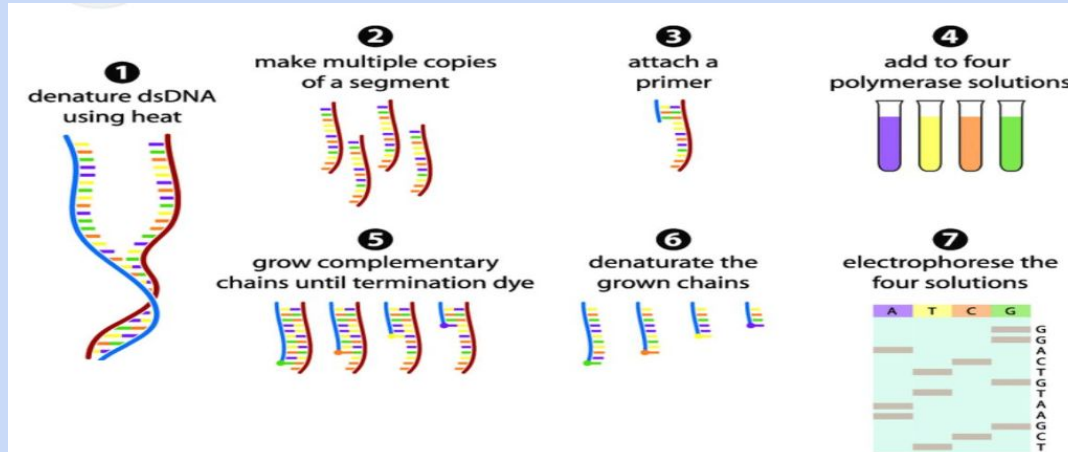


PCR Video

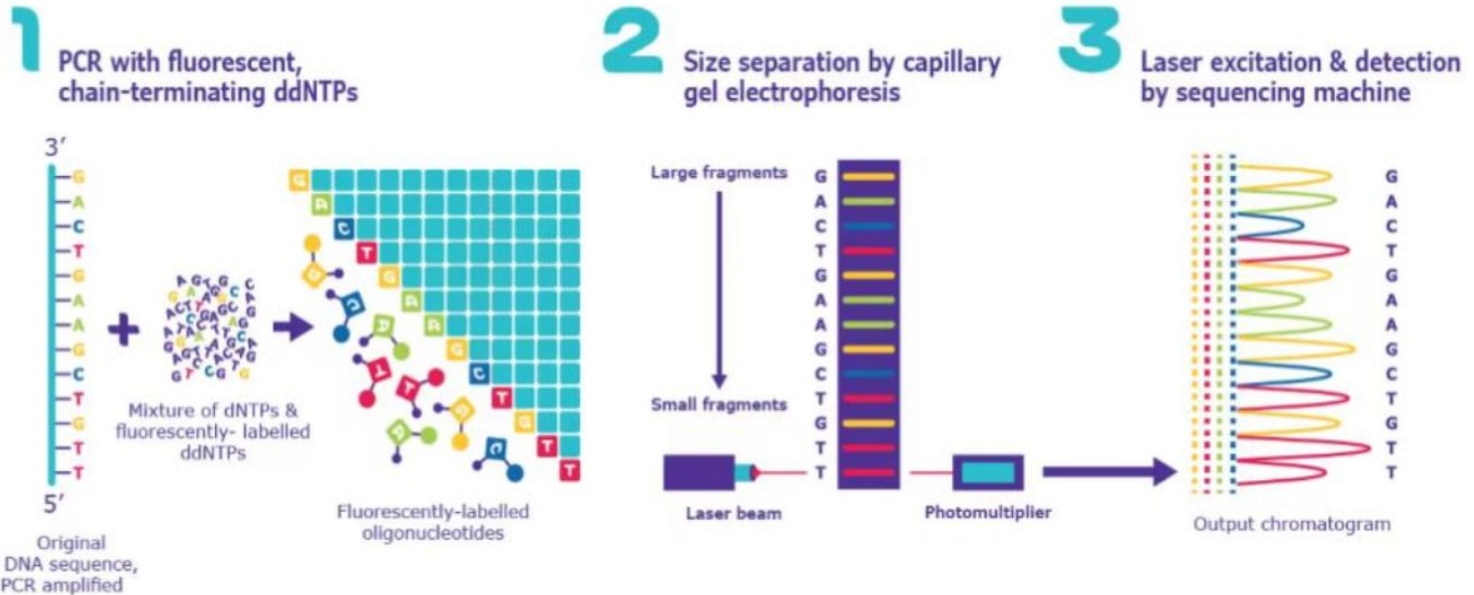
<https://www.youtube.com/watch?v=JRAA4C2OPwg>

What is Sanger Sequencing?

- DNA sequencing, allowing the sequence of nucleotides in DNA to be elucidated
- Utilizes chain-terminating inhibitors called dideoxynucleotides (ddNTPs), which terminate DNA chains at varying bases, enabling DNA fragments of different sizes to be generated



Sanger Sequencing Steps

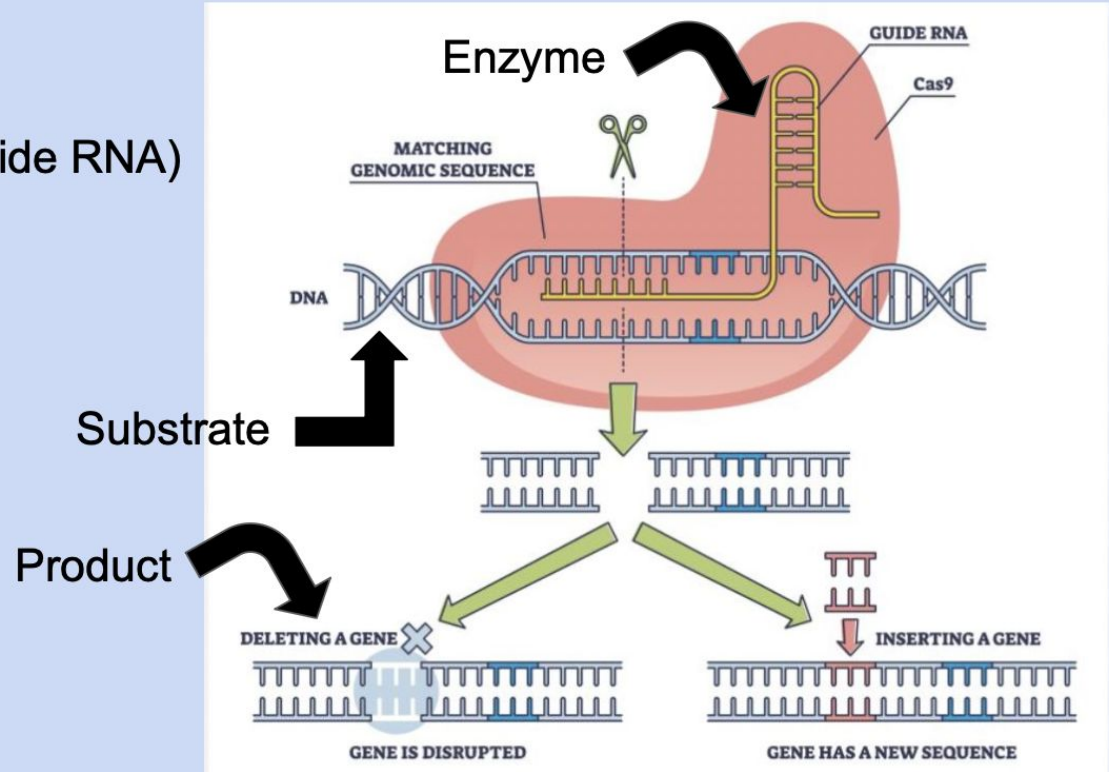


CRISPR-Cas9 Gene Technology

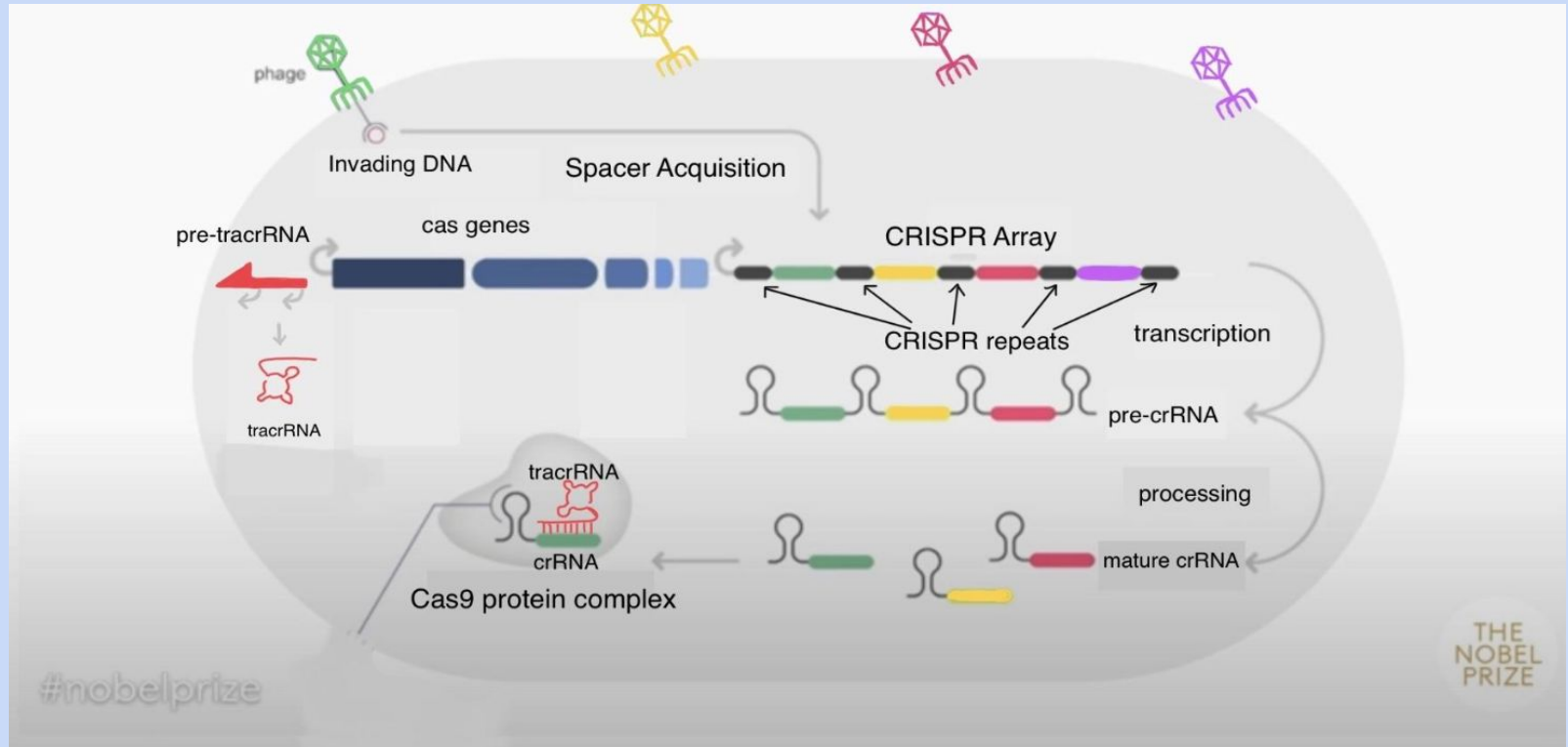
Substrate = DNA Template

Enzyme = Cas Enzyme (+ Guide RNA)

Product = Edited DNA

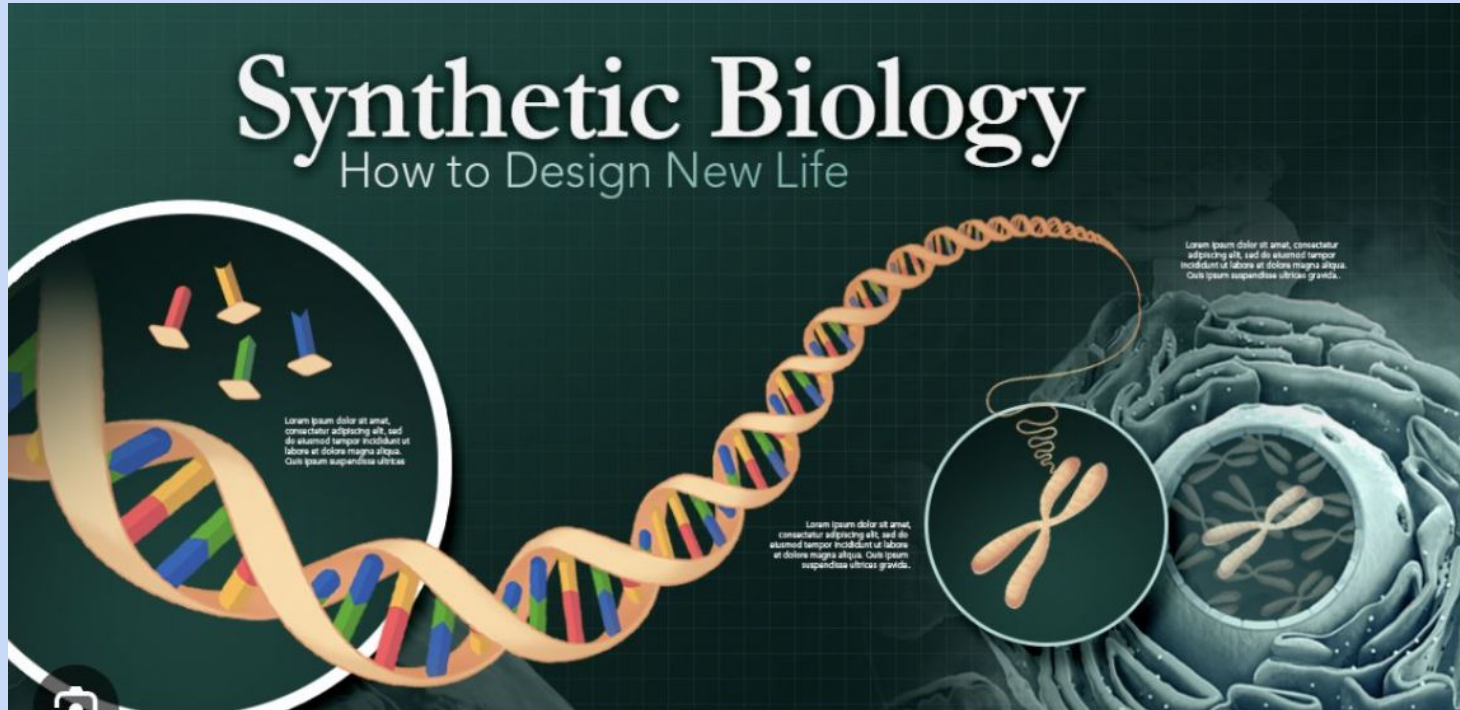


How does CRISPR-Cas9 Mimic Nature?



Nobel Laureates Emmanuelle Charpentier and Jennifer Doudna,
Winners of 2020 Nobel Prize in Chemistry
for CRISPR Gene Editing





We hope you enjoy the course!